CRAWL-WALK-RUN · PART 2

Artificial Intelligence (AI) in the Enterprise: Three Foundational Use Cases Shaping the Next Decade of Work

Ashwin Rangan

Artificial intelligence (AI) has moved from experimental pilots to core enterprise infrastructure. Across industries, boards and executive teams increasingly recognize that AI is not a single technology but a stack of capabilities that reshape how organizations can potentially operate, compete, and create value. However, between potential and realized value is the "Valley of Mud"—where many (most?) enterprises are still muddling. A few have successfully bridged across this valley.

Among these successful enterprises, there is a recurring pattern of emerging applications. Three use cases are surfacing in that pattern: **Conversational Generative AI**, **Coding with Generative AI**, and **Agentic AI**.

Together, these represent a progression—from assisting humans, to augmenting technical work, to autonomously executing business processes. At this stage in both the evolution of AI and its adoption in enterprises, understanding these three use cases is essential for leaders responsible for strategy, governance, and long-term value creation.

Case 1: Conversational Generative AI: The New Enterprise Interface

Conversational Generative AI is the most visible and widely adopted form of AI in the enterprise. It transforms how employees, customers, and partners interact with the unstructured information in the enterprise. At its core, conversational AI uses large language models (LLMs) to understand natural language in the prompting interface, generate responses, and perform tasks through dialogue.

Why it matters: Conversational AI collapses complexity. Instead of navigating multiple applications, employees—using the prompting interface—can ask questions, retrieve data, generate content, and complete tasks through a single interface. This shift mirrors the transition from command-line interfaces to graphical interfaces—but now toward *language* as the universal interface.

Example enterprise applications: Within easy reach of all employees are (1) Knowledge management, wherein AI copilots search and synthesize documents, policies, and data across the enterprise; (2) Employee productivity, where staff are drafting emails, creating presentations, summarizing meetings, and generating reports; and (3) Simple decision support, wherein prompt engineering is generating and providing contextual insights, scenario analysis, and risk considerations.

Strategic impact: Conversational AI democratizes expertise. Employees with varying levels of technical skill gain access to insights previously locked behind specialized tools or expert knowledge. This accelerates decision-making, reduces operational friction, and enhances organizational agility.

Governance considerations: Boards and executive teams must ensure that conversational AI systems are accurate, secure, and aligned with enterprise policies. Key oversight should include probing questions. Executives should confirm that sensitive data is protected during AI interactions, guardrails prevent hallucinations or misinformation, outputs are validated with a human in the loop before use in critical workflows, and training is provided to employees to ensure responsible use.

Conversational AI is the gateway to enterprise-wide AI adoption. It builds literacy, trust, and familiarity—setting the stage for deeper transformation.

Case 2: Accelerating Coding, Software Development and Modernization

The second major use case—Coding with Generative AI—targets software engineering capacity. Generative AI models trained on code can now write functions, generate tests, refactor legacy systems, and assist developers throughout the software development lifecycle.

Why it matters: Most enterprises face a backlog of modernization needs—cloud migration, API development, cybersecurity enhancements, and integration of new digital capabilities. Traditional development cycles are slow, expensive, and dependent on scarce talent. Coding copilots fundamentally change this equation.

Example enterprise applications: The Software Engineering teams can (1) Generate code easily, writing boilerplate code, APIs, scripts, and configuration files; (2) Modernize legacy code by translating COBOL, Java, or .NET systems into modern languages or architectures; and (3) Script quality assurance protocols by generating test cases, detecting vulnerabilities, and improving code reliability.

Strategic impact: Coding with AI increases throughput, reduces defects, and shortens time-to-value. It also enables enterprises to modernize systems that have long been deferred due to cost or complexity. For organizations with aging infrastructure and cumbersome regulatory oversight—enterprises like banks, insurers, utilities, and government agencies—AI-assisted coding may be the only viable path to modernization at scale, with minimized risk.

Governance considerations: Boards must ensure that AI-generated code is secure, compliant, and maintainable. They should ask management: How is intellectual property (IP) risk managed when AI models generate code? What controls ensure that generated code meets enterprise security standards? How are developers trained to collaborate effectively with AI tools? What processes validate AI-generated code before deployment?

Coding with AI is about augmenting Software Engineering teams, not replacing them. Enterprises that adopt AI-assisted development will accelerate modernization and strengthen operational resilience.

Case 3: Autonomous Agentic Systems That Execute Business Processes

Agentic AI represents the third frontier. Unlike conversational AI, which responds to prompts, or coding AI, which assists developers, agentic AI systems can autonomously perform tasks, make decisions within defined boundaries, and orchestrate multi-step workflows across enterprise systems. AI agents combine reasoning, planning, memory, and action. They can analyze data, trigger workflows, interact with applications, and adapt to changing conditions—all with human oversight.

Why it matters: Agentic AI shifts AI from *assistive* to *operational*. It enables enterprises to automate complex processes that previously required routine human judgment and coordination.

Example enterprise applications: For example, (1) Finance could automate autonomous reconciliation, fraud detection, and regulatory reporting; (2) Operations could optimize supply chains, inventory management, and logistics coordination; (3) HR could automate onboarding, compliance checks, and workforce analytics; and (4) IT could automate incident response and continuous monitoring.

Strategic impact: Agentic AI unlocks step-change improvements in efficiency, accuracy, and scalability. It reduces manual workload, minimizes errors, and enables continuous operations. More importantly, it creates new operating models where humans supervise

outcomes while agents execute tasks.

Governance considerations: Agentic AI introduces new oversight responsibilities for boards and management teams. They should debate—for example—how much autonomy agents should have before requiring human review. Or what controls prevent agents from executing unintended actions. And how risks are monitored across autonomous workflows. Or what audit trails should be enabled to ensure accountability and regulatory compliance. And how the enterprise maintains a human-in-the-loop for appropriate governance.

Agentic AI will redefine enterprise operations. It is the most transformative—and the most governance-intensive—of the three use cases.

Conclusion: A Strategic Framework for Enterprise AI Adoption

These three use cases form a maturity curve. Enterprises that adopt all three will create a flywheel of capability with faster development, smarter workflows, and increasingly autonomous systems that drive competitive advantage.

Boards and executives must approach this transformation with both ambition and discipline. The opportunity is immense, but so are the responsibilities. Governance, risk management, ethical principles, and workforce readiness must evolve alongside technological capability.

AI is here to stay. It is the operating system of the now and future enterprise. This is another example of using the crawl-walk-run paradigm to embrace and exploit AI as the new Swiss-Army knife with pluripotent capabilities.